Chapter 10

Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\boldsymbol{A}$

$$\boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

$$\left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

can be written in the more symmetrical form

$$\Box^2 V + \frac{\partial L}{\partial t} = -\frac{1}{\epsilon_0}\rho$$

$$\Box^2\boldsymbol{A} - \boldsymbol{\nabla}L = -\mu_0\boldsymbol{J}$$

where

$$\Box^2 \equiv \boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\Box^2$ and L into the equations for V:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)V + \frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \boldsymbol{\nabla}^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) + \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} =$$

$$= \boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

and for $\boldsymbol{A}$:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

□

Problem 10.2

Problem 10.3

(a) Find the fields, and the charge and current distributions, corresponding to

$$V(\boldsymbol{r},t) = 0, \quad \boldsymbol{A}(\boldsymbol{r},t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}}$$

(b) Use the gauge function $\lambda = -(1/4\pi\epsilon_0)(qt/r)$ to transform the potentials, and comment on the result.

Solution:

The **E** and **B** fields are

$$\mathbf{E} = -\boldsymbol{\nabla}V - \frac{\partial\mathbf{A}}{\partial t}$$
$$\mathbf{B} = \boldsymbol{\nabla}\times\mathbf{A}$$

The derivative of **A** with respect to time is

$$\frac{\partial\mathbf{A}}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}},$$

while the curl of **A** in spherical coordinates is

$$\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta A_\phi) - \frac{\partial A_\theta}{\partial\phi}\right]\hat{\boldsymbol{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(rv_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]\hat{\boldsymbol{\phi}}.$$

Since $A_\phi = A_\theta = 0$, and $\frac{\partial A_r}{\partial\phi} = \frac{\partial A_r}{\partial\theta} = 0$, the curl of **A** is zero.

(a)

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$
$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} = 0$$

(b) Using the gauge function, V and $\mathbf{A}$ are transformed into

$$V'(\mathbf{r}, t) = V(\mathbf{r}, t) - \frac{\partial \lambda}{\partial t}$$
$$\mathbf{A}'(\mathbf{r}, t) = \mathbf{A}(\mathbf{r}, t) + \boldsymbol{\nabla}\lambda$$

Writing out the derivative and gradient of λ

$$\frac{\partial \lambda}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r}$$
$$\boldsymbol{\nabla}\lambda = \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}}$$

gives the transformed potentials

$$V'(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$
$$\mathbf{A}'(\mathbf{r}, t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}} + \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2} = 0$$

$\mathbf{E}$ is now given by $-\boldsymbol{\nabla}V$ (instead of by $-\frac{\partial \mathbf{A}}{\partial t}$), while $\mathbf{B}$ is again 0, this time directly because $\boldsymbol{\nabla} \times \mathbf{A} = \boldsymbol{\nabla} \times 0 = 0$.

$$\mathbf{E} = -\boldsymbol{\nabla}V = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$
$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} = 0$$

The result is exactly the same as in (a).

□

Problem 10.4

Suppose $V = 0$ and $\mathbf{A} = A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$, where $A_0, \omega,$ and k are constants. Find $\mathbf{E}$ and $\mathbf{B}$, and check that they satisfy Maxwell's equations in vacuum. What conditions must you impose on ω and k?

Solution:

Maxwell's equations in vacuum are

$$\text{(i)}\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho \qquad \text{(iii)}\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\text{(ii)}\nabla \cdot \mathbf{B} = 0 \qquad \text{(iv)}\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

where as before we set

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$
$$\mathbf{B} = \nabla \times \mathbf{A}$$

This gives

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = -\omega A_0 \cos(kx - \omega t)\hat{\boldsymbol{y}}$$
$$\mathbf{B} = \frac{\partial A_y}{\partial x}\hat{\boldsymbol{z}} = kA_0 \cos(kx - \omega t)\hat{\boldsymbol{z}}$$

Since $\rho = 0$, this directly gives $\nabla \cdot \mathbf{E} = 0$. (ii) gives $\nabla \cdot \mathbf{B} = 0$.

From (iv) we have the relation, since the current density $\mathbf{J} = 0$,

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$
$$\downarrow$$
$$\nabla \times \mathbf{B} = -\frac{\partial B_z}{\partial x}\hat{\boldsymbol{y}} = k^2 A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$$
$$\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \epsilon_0 \omega^2 A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$$

From this we see that $k^2 = \mu_0 \epsilon_0 \omega^2 \rightarrow k = \sqrt{\mu_0 \epsilon_0}\omega \rightarrow k = \frac{\omega}{c}$

From (iii) we have the relation

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$
$$\downarrow$$
$$\nabla \times \mathbf{E} = \frac{\partial E_y}{\partial x}\hat{\boldsymbol{z}} = \omega k A_0 \sin(kx - \omega t)\hat{\boldsymbol{z}}$$
$$-\frac{\partial \mathbf{B}}{\partial t} = \omega k A_0 \sin(kx - \omega t)\hat{\boldsymbol{z}}$$

Equations (i)-(iv) are all verified with the condition $k = \frac{\omega}{c}$.

□

Problem 10.5

Which of the potentials

$$\text{(i) } V = 0, \quad \mathbf{A} = \begin{cases} \frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\boldsymbol{z}}, & \text{for } |x| < ct \\ 0 & \text{for } |x| > ct \end{cases}$$

$$\text{(ii) } V = 0, \quad \mathbf{A} = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}}$$

$$\text{(iii) } V = 0, \quad \mathbf{A} = A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$$

are in the Coulomb gauge? Which are in the Lorenz gauge?

Solution:

The Coulomb gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = 0$$

and the Lorenz gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Starting with checking the Coulomb gauge, we get

$$\text{(i) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(\frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\boldsymbol{z}} \right) = 0$$

$$\text{(ii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(-\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}} \right) = -\frac{qt}{4\pi\epsilon_0} \boldsymbol{\nabla} \cdot \left(\frac{\hat{\boldsymbol{r}}}{r^2} \right) = -\frac{qt}{4\pi\epsilon_0}\delta^3(\mathbf{r})$$

$$\text{(iii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot (A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}) = 0$$

(i) and (iii) are in the Coulomb gauge, but not (ii). Next, checking Lorenz gauge we see that the divergence of $\mathbf{A}$ should be proportional to the time derivative of V, but since $V = 0$ for (i) and (iii), they are also in the Lorenz gauge. This does not apply to (ii) since the divergence of $\mathbf{A} \neq 0$ while $\frac{\partial V}{\partial t} = 0$.

(i) Both Coulomb and Lorenz.

(ii) Neither Coulomb or Lorenz.

(iii) Both Coulomb and Lorenz. □

Problem 10.6

Problem 10.7

Problem 10.8

Problem 10.9

Problem 10.10

Confirm that the retarded potentials satisfy the Lorenz gauge condition. [Hint: First show that

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath} \right) = \frac{1}{\imath} (\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath} (\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath} \right)$$

where $\boldsymbol{\nabla}$ denotes derivatives with respect to $\mathbf{r}$, and $\boldsymbol{\nabla}'$ denotes derivatives with respect to $\mathbf{r}'$. Next, noting that $\mathbf{J}(\mathbf{r}', t - \imath/c)$ depends on $\mathbf{r}'$ both explicitly and through $\imath$, whereas it depends only on $\mathbf{r}$ through $\imath$, confirm that

$$\boldsymbol{\nabla} \cdot \mathbf{J} = -\frac{1}{c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath), \boldsymbol{\nabla}' \cdot \mathbf{J} = -\dot{\rho} - \frac{1}{c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath)$$

Use this to calculate the divergence of $\mathbf{A}$ (Eq. 10.26.)]

Solution:

We want to show

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

where

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{\imath} \, \mathrm{d}\tau'$$

Now, taking the divergence of $\mathbf{A}$ gives

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}(\mathbf{r}', t_r)}{\imath} \right) \mathrm{d}\tau'$$

From now on, $\mathbf{J}(\mathbf{r}', t_r) = \mathbf{J}$. Using a divergence identity gives

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath} \right) = \frac{1}{\imath} (\boldsymbol{\nabla} \cdot \mathbf{J}) + \mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath} \right)$$

and

$$\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) + \mathbf{J} \cdot \left(\boldsymbol{\nabla}' \frac{1}{\imath}\right)$$

Since $\frac{1}{\imath} = \frac{1}{|\mathbf{r}-\mathbf{r}'|}$, we have

$$\boldsymbol{\nabla}' \frac{1}{\imath} = -\boldsymbol{\nabla} \frac{1}{\imath}$$

which means we can rewrite $\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)$ as

$$\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath}\right)$$

$$\downarrow$$

$$\mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)$$

Putting this expression into $\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath}\right)$ gives

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)$$

which is what the hint wanted us to show. Putting this into the expression for $\boldsymbol{\nabla} \cdot \mathbf{A}$ gives

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \left[\frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)\right] \mathrm{d}\tau' \tag{1}$$

According to the Lorenz gauge condition,

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t} = -\frac{\mu_0}{4\pi} \int \frac{\dot{\rho}}{\imath} \, \mathrm{d}\tau'$$

With this information, we now evaluate the three terms in Equation 1

Starting with the first term, and using the chain rule on $\mathbf{J}$. Since $\mathbf{J}$ depends on $\mathbf{r}$ only through t_r,

$$\boldsymbol{\nabla} \cdot \mathbf{J} = \dot{\mathbf{J}} \cdot \boldsymbol{\nabla} t_r = -\frac{1}{c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath)$$

$$\downarrow \tag{2}$$

$$\frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) = -\frac{1}{\imath c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath)$$

The next term is the divergence of $\mathbf{J}$ with respect to $\mathbf{r}'$. $\mathbf{J}$ depends on $\mathbf{r}'$ both explicitly and through t_r, so

$$\boldsymbol{\nabla}' \cdot \mathbf{J} = (\boldsymbol{\nabla}' \cdot \mathbf{J})_{t_r} + \dot{\mathbf{J}} \cdot \boldsymbol{\nabla}' t_r = -\dot{\rho} - \frac{1}{c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath)$$
$$\downarrow \tag{3}$$
$$\frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) = -\frac{\dot{\rho}}{\imath} - \frac{1}{\imath c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath) = -\frac{\dot{\rho}}{\imath} + \frac{1}{\imath c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath)$$

where $(\boldsymbol{\nabla}' \cdot \mathbf{J})_{t_r}$ is the divergence taken at fixed t_r, which equals $-\dot{\rho}$ by the continuity equation.

The last term integrates to 0 which I will explain later.

Adding Equation 2 and Equation 3, the only term left is

$$-\frac{\dot{\rho}}{\imath}$$

which gives

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\mu_0}{4\pi}\int\left(-\frac{\dot{\rho}}{\imath}\right)\mathrm{d}\tau' = -\frac{\mu_0\epsilon_0}{4\pi\epsilon_0}\int\frac{\dot{\rho}}{\imath}\,\mathrm{d}\tau' =$$
$$= -\mu_0\epsilon_0\frac{\partial}{\partial t}\left[\frac{1}{4\pi\epsilon_0}\int\frac{\rho}{\imath}\,\mathrm{d}\tau'\right] = -\mu_0\epsilon_0\frac{\partial V}{\partial t}$$
$$\downarrow$$
$$\boxed{\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0\epsilon_0\frac{\partial V}{\partial t}}$$

□

Problem 10.11

Problem 10.12

Problem 10.13

Solution:

We have

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}', t_r)}{\imath^2}\hat{\boldsymbol{\imath}} + \frac{\dot{\rho}(\mathbf{r}', t_r)}{c\imath}\hat{\boldsymbol{\imath}} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2\imath}\right]\mathrm{d}\tau' \tag{4}$$

and

$$\rho(\mathbf{r},t) = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)t$$

$$\rho(\mathbf{r},t_r) = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)t_r = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)\left(t - \frac{\mathscr{r}}{c}\right)$$

$$\downarrow$$

$$\rho(\mathbf{r},t_r) = \rho(\mathbf{r},t) - \dot{\rho}(\mathbf{r},0)\frac{\mathscr{r}}{c}$$

$$\downarrow$$

$$\frac{\rho(\mathbf{r},t_r)}{\mathscr{r}^2} = \frac{\rho(\mathbf{r},t)}{\mathscr{r}^2} - \frac{\dot{\rho}(\mathbf{r},0)}{c\mathscr{r}}$$

Putting this into Equation 4, and noting that $\dot{\mathbf{J}} = 0$ since $\mathbf{J}$ is constant in time, we get

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}',t)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} - \frac{\dot{\rho}(\mathbf{r}',0)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}} + \frac{\dot{\rho}(\mathbf{r}',t_r)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}}\right]\mathrm{d}\tau'$$

$$\downarrow$$

$\mathbf{J}$ constant in time $\leftrightarrow$ $\dot{\rho}$ constant in time $[\boldsymbol{\nabla}\cdot\mathbf{J} = -\dot{\rho}] \rightarrow \dot{\rho}(\mathbf{r},t) = \dot{\rho}(\mathbf{r},0)$

$$\downarrow$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}',t)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} - \frac{\dot{\rho}(\mathbf{r}',0)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}} + \frac{\dot{\rho}(\mathbf{r}',0)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}}\right]\mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\frac{\rho(\mathbf{r}',t)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}\,\mathrm{d}\tau'$$

□

Problem 10.14

Show that the Biot-Savart law holds, with $\mathbf{J}$ evaluated at the nonretarded time, considering only the first two terms in the Taylor expansion for $\mathbf{J}$ is

$$\mathbf{J}(t_r) = \mathbf{J}(t) + (t_r - t)\dot{\mathbf{J}}(t) + ...$$

where the $\mathbf{r}$ dependence in $\mathbf{J}(\mathbf{r},t_r)$ has been omitted for compactness.

Solution:

$$\mathbf{B}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int\left[\frac{\mathbf{J}(\mathbf{r}',t_r)}{\mathscr{r}^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}',t_r)}{c\mathscr{r}}\right]\times\hat{\boldsymbol{\mathscr{r}}}\,\mathrm{d}\tau' \tag{5}$$

We use the given Taylor expansion on the first term in Equation 5:

$$\mathbf{J}(t_r) = \mathbf{J}(t) + (t_r - t)\dot{\mathbf{J}}(t)$$

$$t_r = t - \frac{\mathscr{r}}{c} \rightarrow t_r - t = -\frac{\mathscr{r}}{c}$$

$$\downarrow$$

$$\mathbf{J}(t_r) = \mathbf{J}(t) - \frac{\mathscr{r}}{c}\dot{\mathbf{J}}(t)$$

$$\downarrow$$

$$\frac{\mathbf{J}(t_r)}{\mathscr{r}^2} = \frac{\mathbf{J}(t)}{\mathscr{r}^2} - \frac{\dot{\mathbf{J}}(t)}{c\mathscr{r}} \tag{6}$$

and on the second term of Equation 5:

$$\dot{\mathbf{J}}(t_r) = \dot{\mathbf{J}}(t) + (t_r - t)\ddot{\mathbf{J}}(t)$$

But since all higher order terms are neglected, $\ddot{\mathbf{J}}(t) = 0$ so

$$\dot{\mathbf{J}}(t_r) = \dot{\mathbf{J}}(t) \tag{7}$$

Putting Equation 6 and Equation 7 into Equation 5 gives

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{r}', t)}{\mathscr{r}^2} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t)}{c\mathscr{r}} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t)}{c\mathscr{r}} \right] \times \hat{\boldsymbol{\mathscr{r}}}\, \mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t)}{\mathscr{r}^2} \times \hat{\boldsymbol{\mathscr{r}}}\, \mathrm{d}\tau'$$

□

Problem 10.15

Problem 10.16

Problem 10.17

Problem 10.18

Problem 10.19

Problem 10.20

Problem 10.21

Problem 10.22

Problem 10.23

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Problem 10.37

Problem 10.38